

A Minimal Spectral Nemesis Square for the Fâ, Square Program

Overview

This note records a concrete 3×3 "spectral square" derived from a classical prime magic square and packages it in the intersection pairing language used in the Fâ, Square program. Its purpose is to provide a tiny, fully explicit test object whose type matches the ingredients of the Fâ, Square intersection theory (additive spectral data, symmetric bilinear form, eigenvalues/eigenvectors) without making any claim about the Riemann Hypothesis.

The construction starts from a known 3×3 additive prime magic square, passes to the logarithmic (spectral) domain, symmetrizes to obtain a real symmetric matrix, and studies its eigenstructure as a toy "intersection form". This aligns with the characteristic 1 and spectral framing in the arithmetic site literature and in the Missing Object over $\hat{\mathbb{A}}_1$ note without interacting with the true Li sequence or the $\hat{\mathbb{A}}_1$ -derived positivity.[^1]

1. Additive prime magic square

The starting point is a minimal 3×3 additive magic square built entirely from distinct prime numbers. Such squares have been studied in the recreational and combinatorial number theory literature and are used in the Nemesis constructions as exponent bases.[^2]

A concrete example is

\$\$

P =

$\begin{pmatrix}$

17 & 89 & 71 \\

113 & 59 & 5 \\

47 & 29 & 101

$\end{pmatrix}$,

\$\$

where every row, column, and the two main diagonals sum to the same additive magic constant 177.[^3] This square appeared in earlier Nemesis work as the exponent lattice for a multiplicative golden ratio square; here it is treated purely additively.

2. Spectral lift via logarithms

To move into the spectral language that underlies the arithmetic site and scaling flow pictures, the entries of the prime square are mapped through the natural logarithm.[^1]

Definition 2.1 (spectral square). The spectral Nemesis square associated to P is the real matrix

\$\$

$S := (\log P)_{ij} := \log P_{ij}.$

\$\$

Numerically,

\$\$

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S \approx
\begin{pmatrix}
2.8332 & 4.4886 & 4.2627\\
4.7274 & 4.0775 & 1.6094\\
3.8501 & 3.3673 & 4.6151
\end{pmatrix}.
$$
[^3]

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The entries of S live in the same additive real line that appears in the characteristic 1 and tropical models: multiplicative structure on primes is translated into additive structure on their logarithms, and the max-plus semantics of the arithmetic site interpret addition and maximum operations on such real weights.[1]

Row and column sums in S reflect products in P : for example, the spectral row sum satisfies

$\sum_j S_{ij} = \log \Big(\prod_j P_{ij} \Big)$, so multiplicative row products become additive spectral aggregates.

3. Symmetric intersection style matrix

The $\widehat{F}$, $\widehat{\square}$ program formulates its positivity problem as a Hodge index condition on a symmetric bilinear form (an intersection pairing) on a 2-dimensional arithmetic surface.[1] To match that type in this toy setting, the spectral square is symmetrized.

Definition 3.1 (intersection matrix). Define

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H := \frac{1}{2}(S + S^{\text{top}}).
Explicitly,
H \approx
\begin{pmatrix}
2.8332 & 4.6080 & 4.0564\\
4.6080 & 4.0775 & 2.4884\\
4.0564 & 2.4884 & 4.6151
\end{pmatrix}.
$$
[^3]

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The matrix H is real symmetric and thus determines a quadratic form

$Q(v) := v^{\text{top}} H v, \text{quad } v \in \mathbb{R}^3$.

This quadratic form is a finite-dimensional analogue of an intersection pairing: it is a symmetric bilinear form on a (very small) real vector space built functorially from arithmetic data. Unlike the genuine intersection form on the conjectural $\widehat{F}$, $\widehat{\square}$, no positivity or negativity claim is made for Q ; it is simply an explicit test object with the correct type.[1]

4. Eigenstructure and toy Hodge signature

The eigenvalues and eigenvectors of H give a miniature version of the signature considerations that appear in the Hodge index theorem for surfaces. A numerical eigen-decomposition yields

λ

$\lambda_1 \approx -1.6746$

$\lambda_2 \approx 1.9214$

$\lambda_3 \approx 11.2791$

with corresponding orthonormal eigenvectors (columns of V) approximately

V

$V \approx$

$\begin{pmatrix}$

$-0.8023 & -0.1068 & -0.5872$

$0.5054 & -0.6450 & -0.5732$

$0.3175 & 0.7567 & -0.5715$

$\end{pmatrix}$.

λ

$[\lambda^3]$

Two features are noteworthy:

- One eigenvalue is large and positive (λ_3), with eigenvector roughly proportional to $(1, 1, 1)$ up to sign. This provides a natural toy "ample" direction, analogous in shape (though not in meaning) to the ample class singled out in Hodge index formulations.
- One eigenvalue is negative (λ_1), and one is small positive (λ_2), so the signature of H is $(2, 1)$ in this 3-dimensional setting.

By contrast, the classical Hodge index theorem for a projective surface asserts that the intersection form has signature $(1, \rho-1)$ on the Néron-Severi space: one positive direction along the ample class and a negative-definite primitive complement.^[1] The toy spectral square is therefore not intended as a literal model of the surface signature, but as a small, completely concrete example where the same kinds of questions (distinguished directions, sign patterns, primitive subspaces) can be asked and answered without any appeal to $\hat{H}^2$ or Li.

5. Nemesis encoding as a representation layer

In the original Nemesis framework, a prime magic square is exponentiated with base ϕ (the golden ratio) to obtain a multiplicative magic square whose entries lie in the algebraic field $\mathbb{Q}(\sqrt{5})$ via Fibonacci power-reduction identities.^[2] The current spectral square can be wrapped in an analogous way, but with a clear separation between representation and operator.

One possible encoding is

λ

$N_{ij} := \phi^{S_{ij}} = e^{(\log \phi) S_{ij}} = P_{ij}^{\log \phi}$.

\$\$

The log map is explicitly invertible:

\$\$

$$S_{\{ij\}} = \log P_{\{ij\}} = \frac{\log N_{\{ij\}}}{\log \phi}.$$

\$\$

Thus all spectral and intersection questions are properly phrased at the \$\$\$\$ and \$\$H\$\$ level; \$\$N\$\$ is a rigid geometric reexpression of the same data. This matches the representationâ€‘versusâ€‘proof boundary emphasized in the UOR and Fâ, â€‘Square stacks: sophisticated encodings (via contentâ€‘addressing, exceptional groups, or Nemesisâ€‘style exponentiation) organize and transport arithmetic information but do not, by themselves, establish the missing positivity.[^4][^1]

6. Alignment with the Fâ, â€‘Square program

The Fâ, â€‘Square repository formalizes a target object and mechanism: a 2â€‘dimensional arithmetic surface over a characteristicâ€‘1 base, equipped with an intersection pairing and an ample class, whose Hodge index theorem would be equivalent to multiple faces of the Riemann Hypothesis (Weil functional positivity, Li coefficient positivity, cancellation of the prime oscillation, spectral confinement).[^1]

Within that framework, the 3Ã—3 spectral Nemesis square and its intersection matrix \$\$H\$\$ provide:

- A **typeâ€‘matched toy**: a finite real vector space, a symmetric bilinear form extracted from arithmetic data, and a numerically computable signature.
- A concrete sandbox for testing language and intuition about ample directions, primitive complements, and signatures, without any entanglement with $\hat{\eta}$ or the true Li sequence.
- A natural anchor for future experiments in spectral number theory: higherâ€‘order prime squares, alternative symmetric constructions, or Liâ€‘inspired Hankel forms could be compared numerically to the behaviour of \$\$H\$\$ and placed against the templates in ``F1Square.lean``.

This alignment is conceptual rather than evidential: the toy square does not approximate the true Fâ, â€‘square nor suggest any positivity beyond its own finite eigenstructure. Its value is as a minimal, transparent example at the intersection of prime combinatorics, logarithmic spectra, and intersectionâ€‘form language, suitable for use as a worked example or test case in the broader Fâ, â€‘Square program.

References

1.

[missing_object_over_Q.md](https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/141823910/c2d27e66-8fea-4e41-b15d-b1e870b9571a/missing_object_over_Q.md) - Status and stance. This document formalizes the missing object whose absence over is the content of ...

2.

[Nemesis-Square-and-Nemesis-Hyper-Matrix-Multiplicative-Magic-Structures-via-Golden-Ratio-Exponentiation-of-Prime-Sequences.pdf](https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/141823910/e5be4317-f4d4-4918-a23d-20c4b542d6fd/Nemesis-Square-and-Nemesis-Hyper-Matrix-Multiplicative-Magic-Structures-via-Golden-Ratio-Exponentiation-of-Prime-Sequences.pdf)

3. [Li's criterion - Wikipedia](https://en.wikipedia.org/wiki/Li's_criterion)

4.

[singularity_geometry_lens.md](https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/141823910/1d19bb7e-071f-41ac-af05-4309f11026a8/singularity_geometry_lens.md) - Status exposition verified computational toolkit. The organizing lens is classical analytic combinat...